E • FST : 40 • Test ID : 240 • TEST DATE : 12-Mar-2026

NEW LIGHT
INSTITUTE
Medical | Foundation

ALL INDIA
FULL SYLLABUS
TEST SERIES 2025-26

NATIONAL ELIGIBILITY-CUM-ENTRANCE TEST

SOLUTION

PHYSICS

1. (2) [NCERT-XI-I-8]

Let $[G] \propto c^x g^y p^z$

by substituting the following dimensions

$[G] = [M^{-1}L^3T^{-2}]$, $[c] = [LT^{-1}]$

$[g] = [LT^{-2}]$

$[p] = [ML^{-1}T^{-2}]$

and by comparing the powers of both sides we can get

$x = 0, y = 2, z = -1 \quad \therefore \quad [G] \propto c^0 g^2 p^{-1}$

2. (1) [NCERT-XI-I-114]

The M.I of system about x-axis

$= I_1 + I_2 + I_3 + I_4$

$= 0 + 0 + 3 \times (3)^2 + 4 \times (-2)^2$

$= 27 + 16 = 43$ kg-m^2

3. (1) [NCERT-XI-I-134]

$$g = \frac{GM}{R^2},\ V_e = \sqrt{\frac{2GM}{R}} \text{ and } U = -\frac{GMm}{R}$$

$$\therefore\ g \propto \frac{M}{R^2},\ V_e \propto \sqrt{\frac{M}{R}} \text{ and } U \propto \frac{M}{R}$$

If both mass and radius are increased by 0.5% then V_e and U remains unchanged where as g decreases by 0.5%.

4. (4) [NCERT-XI-II-209]

Temperature of mixture

$$\theta = \frac{m_1c_1\theta_1 + m_2c_2\theta_2}{m_1c_1 + m_2c_2}$$

$$32 = \frac{m_1 \times 0.2 \times 40 + 100 \times 0.5 \times 20}{m_1 \times 0.2 + 100 \times 0.5}$$

$m_1 = 375$ gm.

5. (3) [NCERT-XI-II-250]

$v \propto \sqrt{T}$ $\qquad \frac{v_2}{v_1} = \sqrt{\frac{800}{200}} = 2$

6. (2) [NCERT-XII-I-82]

$$t = \frac{1}{258 - 256} = \frac{1}{2} = 0.5 \text{ sec.}$$

7. (3) [NCERT-XII-I-82]

$$V_p - V_q = \left[\frac{6}{3} + \frac{12 \times 6}{12 + 6}\right](0.5)$$

$$= (2 + 4)(0.5) \qquad = 3V$$

8. (4) [NCERT-XII-I-205]

A moving charge and changing electric field both produces magnetic field.

9. (3) [NCERT-XII-I-181]

Phase difference

$$\Delta\phi = \phi_2 - \phi_1 \qquad = \frac{\pi}{6} - \left(-\frac{\pi}{6}\right) = \frac{\pi}{3}$$

10. (1) [NCERT-XII-I-206]

$E\hat{i} \times B\hat{j} = C\hat{k}$

i.e. E is along x-axis and

B is along y-axis.

11. (1) [NCERT-XII-II-240]

For a prism, as the angle of incidence increases, the angle of deviation first decreases, goes to a minimum value and then increases.

12. (2) [NCERT-XII-II-281]

By using $E = W_0 + K_{max}$

$K_{max} = E - W_0$

Hence, $K_1 = 1 - 0.5 = 0.5$

and $K_2 = 2.5 - 0.5 = 2$ $\qquad \frac{K_1}{K_2} = \frac{1}{4}$

13. (2) [NCERT - XI –I–33]

Explanation	Max. value	Min. value
7N and 2 N ⇒	9	5
5N and 3 N ⇒	8	2
10N and 4 N ⇒	14	6

Resultant 9N does not lies b/w maximum 8 and minimum 2 hence option (2) is correct

14. (1) [NCERT - XII –I–118]

For no drift : $V\sin\theta = v$

$$\sin\theta = \frac{v}{V} \therefore t = t_{AB} + t_{BA}$$

$$t = \frac{2\ell}{V\cos\theta} = \frac{2\ell}{V\sqrt{1 - \frac{v^2}{V^2}}} \Rightarrow t = \frac{2\ell}{\sqrt{V^2 - v^2}}$$

15. (4) [NCERT - XI –I–82]

$$\mu_s = \frac{\text{Length of the chain hanging from the tabl}}{\text{Length of the chain lying on the table}}$$

$$\mu_s = \frac{\frac{4}{5}}{1 - \frac{4}{5}} = 4 \qquad \because \mu_k \le \mu_s$$

16. (4) **[NCERT - XII –I–15]**

17. (4) **[NCERT - XII –I–72]**

Charge on 9μF capacitor will be.

Q=cv Q=9x10 Q=90μC

18. (2) **[NCERT - XI –I–61]**

$\varepsilon = Blv$

$= 0.5 \times 2 \times 1 = 1V$

19. (4) **[NCERT - XII –II–266]**

$\beta = \frac{D\lambda}{d}$ $\frac{\beta_2}{\beta_1} = \frac{\lambda_2 D_2 d_1}{\lambda_1 D_1 d_2}$

$\beta_2 = 2.5 \times 10^{-4} m$

20. (2) **[NCERT - XII –II–336]**

$V_{(3k\Omega)} = 30 - 18 = 12$ $i_L = \frac{V_L}{R_L} \Rightarrow \frac{12}{3 \times 10^3}$

$= 4 \times 10^{-3}$ $i = 4$ mA

21. (3) **[NCERT-XI-I-14]**

$\frac{30^2 - 24^2}{24^2 - 0^2} = \frac{2a \times 90}{2a \times s}$

$s = 160m$

22. (1) **[NCERT-XII-I-92]**

$P = \frac{i^2 R}{746} = \frac{(30)^2 \times 15}{746} = 18(HP)$

23. (3) **[NCERT-XI-I-114]**

$I = \frac{2}{5}MR^2$

$\therefore M = \frac{4}{3}\pi R^3 \rho$

$R = \left(\frac{3M}{4\pi\rho}\right)^{\frac{1}{3}}$

$\therefore I = \frac{2}{5}M\left(\frac{3M}{4\pi\rho}\right)^{\frac{2}{3}}$

$= \sqrt[3]{\frac{8}{125}\frac{M^5 \times 9}{16\pi^2\rho^2}}$

$= \sqrt[3]{\frac{9}{250}\frac{M^5}{\pi^2\rho^2}}$

24. (3) **[PYQ Modified]**

25. (1) **[NLI Expert]**

In isothermal process

$(2P_0)V_x = P_0 V_0$

$V_x = V_0/2$

26. (1) **[NLI Expert]**

$\frac{K}{U} = \frac{\frac{1}{2}mw^2(A^2 - y^2)}{\frac{1}{2}mw^2 y^2} = \frac{A^2 - P^2/4}{A^2/4} = \frac{3}{1}$

27. (4) **[NCERT-XII-I-68]**

$i_d = C\frac{dV}{dt}$

28. (2) **[NCERT-XII-II-269]**

29. (3) **[NCERT-XII-II-300]**

$\frac{1}{\lambda} = R\left(\frac{1}{1^2} - \frac{1}{3^2}\right)$

$\frac{1}{\lambda} = R\frac{8}{9}$ $\lambda = \frac{9}{8R}$

30. (2) **[NCERT-XII-II-334]**

$i = \frac{5}{50} = .1A$

31. (1) **(Old NCERT XI Page No. 26)**

$d = \frac{a^{\frac{3}{2}}b^{\frac{1}{4}}}{c}$

$\frac{\Delta d}{d} x100 = \frac{3}{2}\left(\frac{\Delta a}{a} \times 100\right) + \frac{1}{4}\left(\frac{\Delta b}{b} \times 100\right) + \left(\frac{\Delta c}{c} \times 100\right)$

$= \left(\frac{3}{2} \times 2\right) + \frac{1}{4}(4) + 2$

$= 3 + 1 + 2 = 6\%$

32. (4) **[New NCERT XI Page No. 168, 169]**

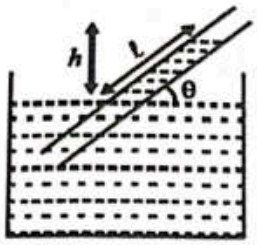
$\text{Stress} = \frac{\text{force}}{\text{area}}$

Both wires have same length and same cross-sectional area, force exerted on both wire will be same, therefore, stress is same for both aluminium and steel wire.

As Young modulus Y of steel wire is more than a aluminium wire. Therefore, strain is different for both wires as,

$\text{Strain} = \frac{\text{Stress}}{Y}$

33. (4) **(New NCERT XI Page No. 196, 197)**



Vertical height of water column remains the same

$l\sin\theta = h$

$l\sin 37^\circ = 9$

$$l = \frac{9}{\frac{3}{5}} = 15 \text{ cm}$$

34. (4) **(New NCERT XII Page No. 310)**

$Q = 8 \times 7.06 - 7 \times 5.60 = 17.3$ MeV

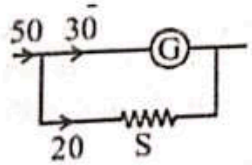
35. (4) **[OLD NCERT XII Page No. 502, 205]**

When both A and B are low then no current flows through resistor and hence $Y = 0$. Now if any of A and B or both (A and B) become high then current starts flowing through resistor and output Y become high. Hence truth table becomes of OR gate.

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

36. (3) **[NCERT-XII-I-186]**

I = for 50 division = 50 (let)
then I_g = for 30 division = 30
from fig.

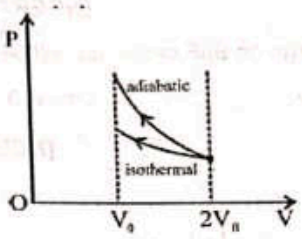


$30\,G = 20\,S$

$30 \times 30 = 20\,S$

$S = 45\Omega$

37. (2) **[NCERT-XI-II-348]**



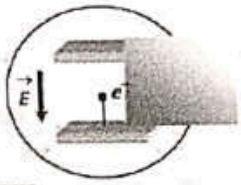
W_{ext} = negative of area with volume - axis
W (adiabatic) > W (isothermal)

38. (1) **[NCERT-XI-I-43]**

$F\cos\theta > \mu(mg - F\sin\theta)$

39. (2) **[NCERT-XII-II-390]**

In electric field photoelectron will experience force and accelerate opposite to the field so it's K.E. increases (i.e. stopping potential will increase), no change in photoelectric current, and threshold wavelength.



40. (3) **[NCERT-XII-I-31]**

$|P_1| = ql$ $|P_2| = ql$

$|\text{Resultant}| = 2P\cos 30^0$

$$2ql\left(\frac{\sqrt{3}}{2}\right) = \sqrt{3}\,ql(-\hat{j}) = -\sqrt{3}ql\hat{j}$$

41. (3) **[NCERT-XI-I-17]**

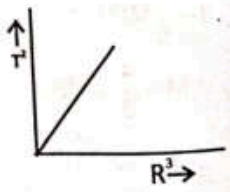
NCERT Theory.

42. (1) **[NCERT-XII-I-16]**

Magnetic force on loop in UMF is always zero.

43. (1) **[PYQ Modified- 20]**

$T^2 \propto r^3$, $T^2 = kr^3$



44. (3) **[PYQ Modified- 20]**

45. (3) **[NCERT-XII-II-]**

$$m_0 = \frac{L}{f_0} = \frac{20}{1} = 20$$

$$m_e = 1 + \frac{D}{f_e} = 1 + \frac{25}{5} = 1 + 5 = 6$$

$$M = m_0 \times m_e = 20 \times 6 = 120$$

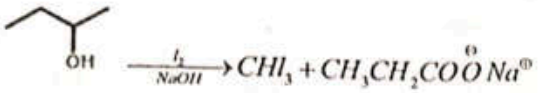
CHEMISTRY

46. (4) [Mod. PYQ]

47. (1) [NCERT-XII-I-132]

$t_{2g}^3 eg^1$

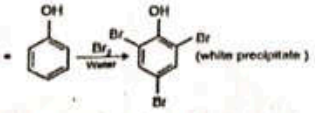
48. (2) [NCERT-XII-240]

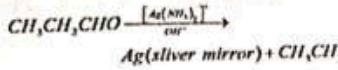


49. (4) [Practical Chemistry]

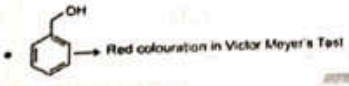
$(CH_3)_3C-OH \xrightarrow[HCl]{ZnCl_2} (CH_3)_3C-Cl$

Immediate sturbidity is because of the formation of $(CH_3)_3C-Cl$





$Ag(sliver\ mirror) + CH_3CH_2COO^-$



[NCERT-XII-293]

50. (1)

In DNA, T joins with A by two hydrogen bonds.

[NCERT-XII-169]

51. (3)

Compounds	B.P. (K)
	453
	446
	446

[NCERT-XII-132]

52. (4)

$\Delta \propto \bar{\upsilon}$

$\bar{\upsilon}_0 = \frac{4}{9}\bar{\upsilon}_t$

[NCERT-XII-5]

53. (2)

$$\frac{X_{Solute}}{X_{Solvent}} = \frac{\frac{n_{Solute}}{n_{solute}+n_{Solvent}}}{\frac{n_{Solvent}}{n_{solute}+n_{solvent}}} = \frac{n_{solute}}{n_{solvent}}$$

$$\text{Molarity} = \frac{number\ of\ moles\ of\ solute}{Volume\ of\ solution\ in\ litres}$$

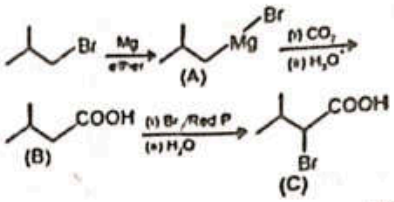
So, option (1) and (4) are correct.

[NCERT-XII-276]

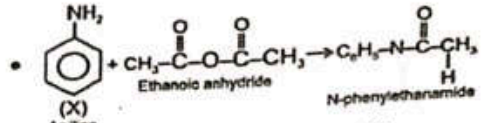
(4)

$PhNH_2 \xrightarrow{HNO_2/0^\circ C} PhN_2^+ \xrightarrow{HBF_4} PhN_2^+(BF_4)^- \xrightarrow{\Delta} Ph-F + N_2 + BF_3$

55. (2) [PYQ Mod.]



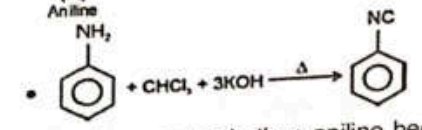
56. (2) [PYQ Mod.]



(X) Aniline

Ethanoic anhydride

N-phenylethanamide



- Pyridine is more basic than aniline because in aniline the lone pair of nitrogen is delocalized.

[PYQ Modified]

57. (3)

An Atom has three fundamental particles - electron, proton and neutron.

The mass of the electron is 9.10939×10^{-31} kg.

Neutrons and protons, both are collectively known as nucleons.

All the isotopes of a given element show same chemical properties.

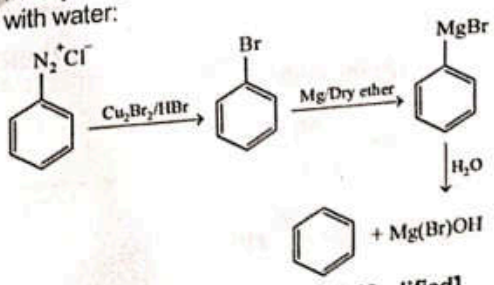
Dalton's atomic theory, regarded atom as an ultimate particle of matter.

[PYQ Modified]

58. (3)

Bromobenzene is formed in first step (Sandmeyer reaction), which further gives phenyl magnesium bromide.

Phenyl magnesium bromide further gives benzene with water:



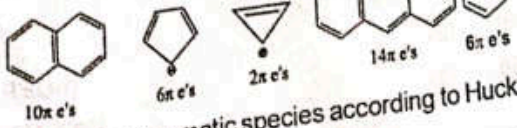
[PYQ Modified]

59. (3)

According to Huckel rule, A compound/molecule is said to be aromatic if it have cyclic conjugate and $[4n + 2]\pi$ electron system.

$\Rightarrow$ Molecule must be sp^2 hybridised

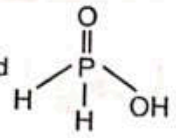
$\Rightarrow$ Molecule must be planar.



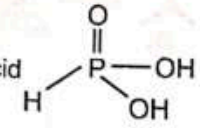
All these are aromatic species according to Huckel rule.

60. (2) **[PYQ Modified]**

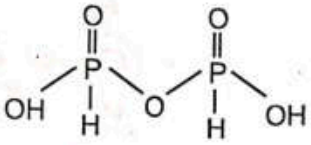
Hypophosphorus Acid



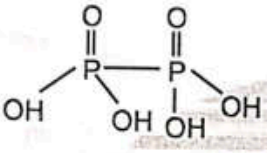
Orthophosphorus Acid



Pyrophosphorus Acid

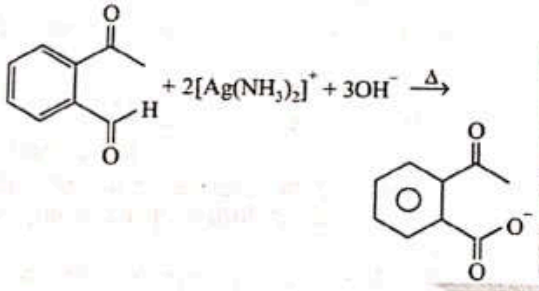


Hypophosphoric Acid



61. (4) **[PYQ Modified]**

Tollen's reagent oxidises aldehydes into carboxylate ion whereas ketone is not oxidised by tollen's reagent.



62. (2) **[NCERT-XII-II-287]**

63. (4) **[NCERT-XI-I-17]**

$\because$ 12 amu=150

$\therefore$ $1amu=\frac{150}{12}$

$\therefore$ $16\ amu=\frac{150}{12}x16$

= 200 unit

64. (1) **[NCERT-XII-I-80]**

$\Delta H = Ea(f) - Ea(b)$

= 32 – 58

= 26 KJ mol^{-1}

Exothermic

65. (2) **[NCERT-XII-110]**

66. (3) **[NCERT-XII-II-251]**

67. (1) **[NEW NCERT-XI-I-24]**

$$\text{Molality } (m) = \frac{w_B \times 1000}{M.M \times W}$$

$$b = \frac{c \times 1000}{M.M \times (a-c)}$$

$$M.M = \frac{c \times 1000}{b(a-c)}$$

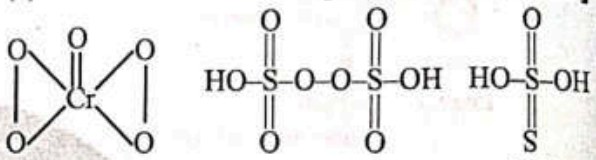
68. (4) **[OLD NCERT-XI,XII-I- 328,174,185]**

69. (3) **[NEW NCERT-XII-II-262]**

Carbylamine on reduction gives 2° amine.

$RNC \xrightarrow{H_2/Ni} R-\underset{(2^\circ\ amine)}{NH}-CH_3$

70. (4) **[OLD NCERT-XII-I-189]**



(I) (II) (III)

So, (I) and (II) have (–O–O–) linkage.

71. (1) **[NEW NCERT-XII-II-251]**

I 2-Chloroethanoic acid is more acidic then ethanoic acid due to –I effect of chlorine.

II As + I of alkyl groups increase, acidic strength decreases.

III alkynes are more acidic than alkenes and alkanes due to sp-hybridisation, of carbon atom which has as electron withdrawing.

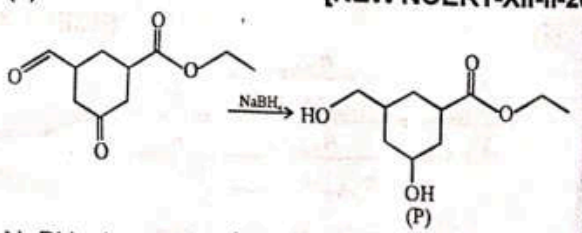
72. (4) **[NEW NCERT-XI-I-89]**

(1) ΔH_3 is more negative than ΔH_1 and ΔH_2 because the electron gain enthalpy of chlorine is highest.

(2) ΔH_1 is less negative than ΔH_2 because electron gain enthalpy of fluorine is more than that of oxygen.

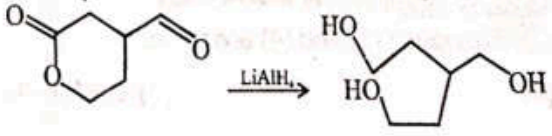
(3) ΔH_4 is positive because electron inters into already half filled stable electronic configuration of nitrogen atom.

73. (4) **[NEW NCERT-XII-II-200]**



(P)

$NaBH_4$ does not reduced ester group.

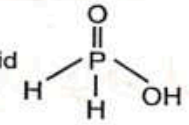


(Q)

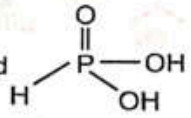
$LiAlH_4$ reduces aldehyde, ketones, carboxylic acid esters to alcohols so, there are five alcoholic groups P and Q.

60. (2) **[PYQ Modified]**

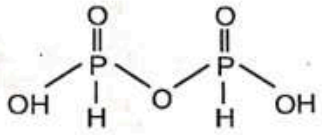
Hypophosphorus Acid



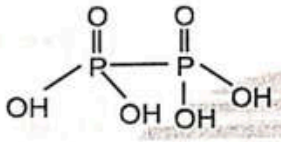
Orthophosphorus Acid



Pyrophosphorus Acid

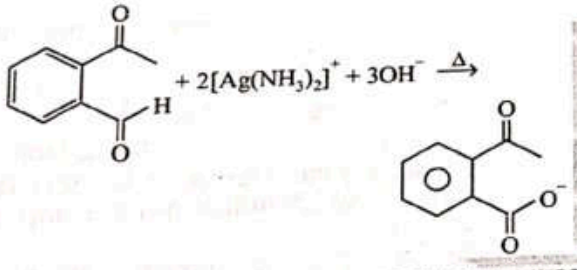


Hypophosphoric Acid



61. (4) **[PYQ Modified]**

Tollen's reagent oxidises aldehydes into carboxylate ion whereas ketone is not oxidised by tollen's reagent.



[NCERT-XII-II-287]

62. (2) **[NCERT-XI-I-17]**

63. (4)

$\because$ 12 amu=150

$\therefore$ $1amu=\frac{150}{12}$

$\therefore$ $16\ amu=\frac{150}{12}\times16$

$= 200$ unit **[NCERT-XII-I-80]**

64. (1)

$\Delta H = Ea(f) - Ea(b)$

$= 32 - 58$

$= 26\ KJ\ mol^{-1}$

Exothermic **[NCERT-XII-110]**

65. (2) **[NCERT-XII-II-251]**

66. (3)

67. (1) **[NEW NCERT-XI-I-24]**

$$\text{Molality } (m) = \frac{w_B \times 1000}{MM \times W}$$

$$b = \frac{c \times 1000}{MM \times (a-c)}$$

$$MM = \frac{c \times 1000}{b(a-c)}$$

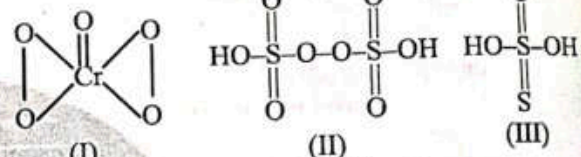
68. (4) **[OLD NCERT-XI,XII-I- 328,174,185]**

69. (3) **[NEW NCERT-XII-II-262]**

Carbylamine on reduction gives 2° amine.

$$RNC \xrightarrow{H_2/Ni} \underset{(2^\circ \text{ amine})}{R-NH-CH_3}$$

70. (4) **[OLD NCERT-XII-I-189]**



(I) (II) (III)

So, (I) and (II) have (–O–O–) linkage.

71. (1) **[NEW NCERT-XII-II-251]**

I 2-Chloroethanoic acid is more acidic then ethan acid due to –I effect of chlorine.

II As + I of alkyl groups increase, acidic strength decreases.

III alkynes are more acidic than alkenes and alka due to sp-hybridisation, of carbon atom which as electron withdrawing.

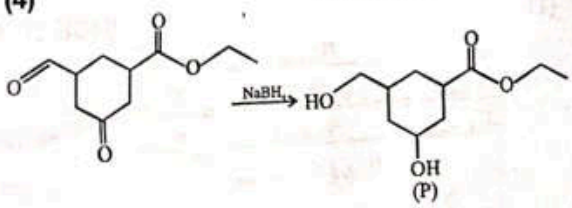
72. (4) **[NEW NCERT-XI-I-89]**

(1) ΔH_3 is more negative than ΔH_1 and ΔH_2 beca the electron gain enthalpy of chlorine is highest.

(2) ΔH_1 is less negative than ΔH_2 because elec gain enthalpy of fluorine is more than that of oxyg

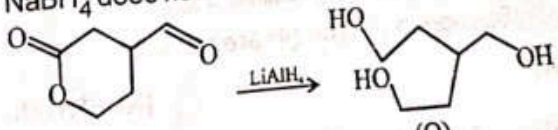
(3) ΔH_4 is positive because electron inters into al half filled stable electronic configuration of nitrogen a

73. (4) **[NEW NCERT-XII-II-20**



(P)

$NaBH_4$ does not reduced ester group.



(Q)

$LiAlH_4$ reduces aldehyde, ketones, carboxylic ac esters to alcohols so, there are five alcoholic g P and Q.

74. (3) [NCERT-XII-9]

For IE_2 Ti^+ V^+ Cr^+ Mn^+

$4s^1$ $4s^1$ $4s^0$ $4s^1$

$3d^2$ $3d^3$ $3d^5$ $3d^5$

IE_2 order = Cr > Mn > V > Ti

75. (4) [NCERT-XI-180]

76. (2) [NCERT-XII-II-270]

77. (4) [NCERT-XII-I-36]

$$E_{cell} = E^{\circ}_{cell} \frac{.059}{4} eg \frac{[Fe^{++}]^2}{Po_2 \times [H^+]^4}$$

$$= 1.67 - \frac{.059}{4} \log \frac{(10^{-3})^2}{(0.1)(10^{-3})^4}$$

$$= 1.67 - \frac{.059}{4} \times 7 = 1.67 - .10$$

$$= 1.57\ V$$

78. (1) [NCERT-XII-I-80]

$$\text{slope} = -\frac{E_a}{R} = -\frac{10}{5}$$

$$= E_a = 2R$$

79. (3) [NCERT-XII-I-106]

80. (3) [NCERT-XI-I-111]

$CHCl_3 < NH_3 < HF < H_2O$ [NCERT-XI-I-162]

81. (2)

$H_2O_{(l)} \rightarrow H_2O_{(g)}$

$\Delta ng = +ve$

$\frac{\Delta S = +ve}{system}$ $\Delta S_{surr} = -ve$

82. (4) [NMC SYLLABUS]

At pH 9-11, dimethylglyoxime forms a red precipitate with Ni^{2+} a Ni-DMG complex.

83. (4) [NMC SYLLABUS]

When H_2S is passed through a mixture containing Cu^{+2}, Ni^{+2}, Zn^{+2} in acidic solution then ion will precipitate Cu^{+2}

84. (1) [Practical Chemistry]

85. (3) [NCERT-XII-6]

Higher the value of Henry's law constant, lower is solubility of gas in liquid.

Correct order of solubility is $He < H_2 < O_2$

86. (4) [NCERT-XII-110]

Mischmetall consist of a lanthanoid metal (95%) and iron (5%) and traces of S, C, Ca and Al.

87. (2) [NCERT-XII-103]

V_2O_5 – Contact process

Finly divided iron – Haber process

Ni – Hydrogenation process

Iron(III) – $S_2O_8^{2-} \rightarrow 2SO_4^{2-}$

88. (2) [Mod. PYQ]

II > I > IV > III

89. (2) [Mod. PYQ]

$x\left(\lambda_m^{\infty}\right)_{A^{x+}} + y\left(\lambda_m^{\infty}\right)_{B^{x-}}$

90. (3) [Mod. PYQ]

Assertion is true but Reason is false.

BIOLOGY

91. (3) [NCERT-I-122 to 127]

92. (1) [NCERT-I-6 to 8]
The lowest taxonomic category that is common to both potato and brinjal is Genus (*Solanum*)

93. (2) [NCERT-I-60]
Pulvinus is
(2) Swollen leaf base in some leguminous plants

94. (4) [Old-NCERT-I]
Tracheids are found in
(4) All categories of vascular plants

95. (3) [NCERT-I-76, 77]
The structure which is present in monocot stem is
(3) Sclerenchymatous hypodermis

96. (3) [NCERT-I-26 to 28]

97. (3) [NCERT-I-135]

98. (3) [NCERT-I-155 to 163]
(a) The substrate used in protoplasmic respiration may be contain nitrogen.
(b) Facultative anaerobes respire in the absence of oxygen
(c) Substrate level ATP synthesis also occurs during glycolysis
(d) The ultimate product of glycolysis is pyruvic acid

99. (3) [NCERT-I-163, 164]
(1) (i) Carbohydrates, (ii) 1
(2) (i) Proteins, (ii) < 1
(3) (i) Fats, (ii) < 1
(4) (i) Malic acid, (ii) > 1

100. (4) [NCERT-I-175 to 178]
(4) a–(ii), b–(i), c–(iv), d–(iii)

101. (1) [NCERT-I-166]
In sexually reproducing organisms, the vital link between two successive generation is Zygote

102. (3) [NCERT-II-73]

103. (4) [NCERT-II-96]

104. (2) [NCERT-II-219]
Species in Amazonian rainforest
(1) Mammals < Birds
(2) Amphibians > Reptiles
(3) Birds > Reptiles

105. (2) [NCERT-II-120, 121]

106. (3) [NCERT-II-47]
(3) Genital warts – Viral disease

107. (1) [NCERT-I-209 to 212]

108. (2) [NCERT-I-46]
a. Water vascular system
b. Radial symmetry in adults
c. Organ system level of organization
(2) Asterias, Echinus, Ophiura – Echinodermata

109. (2) [NCERT-II-44]
The correct set of barrier methods of contraception
(2) Cervical caps, Vaults, Diaphragms

110. (3) [NCERT-II-120, 121

111. (3) [NCERT-I-97

112. (3) [NCERT-I-121, 12

113. (4) [NCERT-I-7
(4) Absorption of water and minerals – Root hair

114. (2) [NCERT-II-7
Substitution of a purine base with a pyrimidine base and vice versa is called
(2) Transversion

115. (3) [NCERT-II-
Monosomy– Turner's syndrome

116. (2) [NCERT-I-106, 1
(2) $CH_3–(CH_2)_{14}–COOH$: Palmitic acid, a saturated fatty acid

117. (3) [NCERT-I-49,
An important general characteristic that reptiles sh with aves is Oviparity

118. (4) [NCERT-I

119. (3) [NCERT-I

120. (1) [NCERT-II
Recovery of healthy plants from diseased pla possible by Meristem culture

121. (2) [NCERT-I
(2) Statins – Monascus purpureus

122. (2) [NCERT

123. (4) [NCERT-II-135
Statement A : Histamine is a potent vas released from mast cells during an allergic re
Statement B : Memory based acquired r evolved in vertebrates.

124. (4) [NCERT-II-1

	(i)	(ii)	(iii)	(iv)
(4)	q	s	t	p

125. (1) [NCERT-II-71]

126. (3) [Old NCERT-I]

(3)  – Present in heart

127. (4) [NCERT-I-91 to 100]

(4) Some organelles are single membrane bound present in eukaryotic cell.

128. (3) [NCERT-I-194 to 196]

(3) O^- blood group — Antibodies A and B present

129. (3) [NCERT-I-224 to 226]

130. (1) [NCERT-I-43 to 48]

	Animal	Phylum	Features
(1)	Clarias	Chordata	Cycloid scales, Operculum

131. (1) [NCERT-II-130 to 133]

	Disease	Pathogen	Symptoms
(1)	Amoebiasis	Entamoeba histolytica	Internal bleeding Blockage of intestinal passage

132. (2) [NCERT-I-143 to 148]

(2) a(ii), b(iii), c(iv), d(i)

133. (3) [NCERT-II-137]

134. (1) [NCERT-II-179,180]

'Cry' proteins. – They are solubilized in alkaline environment of insect midgut

135. (4) [Old-NCERT-I]

(4) Phallomere is present only in male cockroach

136. (1) [NCERT-II-96]

137. (3) [NCERT-II-90, 91]

138. (1) [NCERT-II-33]

During oogenesis in humans, second meiotic division is completed

a. After puberty

b. After ovulation

c. Before fertilization

139. (2) [NCERT-II-28 to 31]

140. (1) [NCERT-II-32]

141. (4) [NCERT-I-232, 233]

(4) a(iv), b(i), c(ii), d(iii)

142. (3) [NCERT-I-248, 249]

Steroid hormones

...through receptors present within the cell

143. (4) [NCERT-II-197 to 202]

(4) Herbivores are considered as a predator.

144. (4) [NCERT-I-247 to 249]

(4) Hormones which interact with intracellular receptors do not generate second messengers.

145. (3) [Old NCERT-I-]

(A) Cockroach consists of a series of fused, segmentally arranged ganglia

(B) Ganglia joined in cockroach by paired longitudinal connectives on the ventral side

(C) Nervous system present on ventral side of cockroach

(D) In cockroach, total no. of ganglia is Nine in number.

146. (3) [NCERT-I-101, 102]

	Meta centric	submeta centric	Acro centric	Telo centric
(3)	(i)	(iv)	(iii)	(ii)

147. (3) [NCERT-II-168 to 170]

When foreign DNA is inserted into any vector, it results in inactivation of any marker gene. This is used for selection of

(3) Recombinant cells

148. (4) [NCERT-I-105, 106]

The correct about amino acid?

(1) It is amphoteric

(2) Present as zwitter ion at isoelectric point

(3) It is unit of protein

149. (3) [NCERT-I-245]

Statement-1 : Cortisol hormone helps in erythropoiesis

Statement-2 : Cortisol hormone increases BMR

150. (2) [NCERT-I-59 to 61]

Statement-I : Leaves originate from ...ot apical meristems

Statement-II : Leaves are arranged in acropetal order

151. (4) [NCERT-II-218]

(4) A-I, B-ii, C-iii, D-iv

152. (2) [NCERT-II-95, 96]

The chemical method developed by Har Gobind Khorana that was instrumental in synthesising RNA molecules with defined combinations of nitrogen bases

153. (3) [NCERT-II-174]

	A	B	C	D
(3)	Motor	Foam breaker	Impeller	Culture broth

154. (2) [NCERT-II-172 to 173]

155. (3) [NCERT-II-209]

156. (1) [NCERT-II-224, 225]

157. (1) [NCERT-II-82]

(1) Both A and R are true and R is the correct explanation of A.

158. (3) [NCERT-II-166, 167]

159. (3) [NCERT-II-135, 136]

160. (1) [NCERT-II-18]

161. (1) [NCERT-II-13]

(1) Both (A) and (R) are true and (R) explains (A) correctly.

162. (1) [NCERT-II-22-23]

(1) Both (A) and (R) are true and (R) explains (A) correctly.

163. (1) [NCERT-II-157]

(1) Both (A) and (R) are true and (R) explains (A) correctly.

164. (3) [NCERT-II-5 to 13]

(3) B, A, C, D, E

165. (2) [NCERT-I-19]

Animal and protozoa – Mode of nutrition is holozoic

(2) Panthera

166. (4) [NCERT-I-21]

(4) Viroids – Low molecular weight of RNA

167. (3) [NCERT–II–197 to 202]

Statement I : Species 'A' Release a chemical which Promote the growth of species 'B'. Such interaction are called Commensalism

Statement II : Species 'C' takes Nutrition and also live on the species 'D'. Such Interaction are called Parasitism

168. (1) [NCERT–II–194]

(1) b - *Per capita* birth rate

d - *Per capita* Death rate

169. (3) [NCERT-I-24 to 32]

Statement I : In Polytrichum and Adiantum the Reduction Division takes place in spore mother cell.

Statement II : In Most Algae the Reduction division takes place in Zygote.

170. (1) [NCERT-I-99 to 100]

The correct statement for centrosome

A. A centrosome are surrounded by amorphous pericentriolar materials

B. They are made up of nine evenly spaced peripheral Fibrils of tubulin Protein

C. Each of peripheral fibrils is a triplet

D. The adjacent triplets are also linked

171. (4) [NCERT-II-183,

Transgenic Animals are useful in many Aspects. t

Aspects are

A. Normal physiology and Development

B. Study of Disease

C. Biological products

D. Vaccine safety

E. Chemical safety testing

172. (3) [NCERT-II-34,

173. (1) [NCERT-I-42 to

Statement I:

Open circulatory system present in, Ascidia and Sq and closed circulatory system in earthworm.

Statement II :

External fertilisation occurs in Nereis, earthworm a pigeon internal fertilisation occurs

174. (2) [NCERT-I-82 to

In Frog what is the ratio of number of Testis, kidn urinary bladder, cloaca respectively.

(2) 2 : 2 : 1 : 1

175. (2) [NCERT-II-

Statement I:

Specific Bt toxin genes were isolated from Bac thuringiensis and incorporated into the several c plants such as cotton.

Statement II :

The choice of genes depends upon the crop and targeted pests as most Bt toxins are Insect-sp specific.

176. (3) [NCERT-II-154

The component of biogas

(A) H_2S

(B) CH_4

(C) CO_2

177. (1) [NCERT-

RNA interference a novel strategy are applied on T plants against Nematode

178. (1) [NCER

(1) Calcium > Sulphur > Magnesium

179. (4) [NCERT-

(4) Pinus < Selaginella < Marchantia < Lamn

180. (2) [NCER

(2) B, A, C